QUERIES :

1.Create Database PCCOE

use PCCOE

```
> use PCCOE
switched to db PCCOE
```

2. Create following Collections

Teachers(Tname,dno,dname,experience,salary,date_of_joining)

Students(Sname,roll_no,class)

db.Teachers.insertMany([{Tname: "Praveen", dno: 1, dname: "COMP", experience: 5, salary: 12000}, {Tname: "Amit", dno: 2, dname: "IT", experience: 3, salary: 9000}, {Tname: "Sneha", dno: 3, dname: "E&TC", experience: 7, salary: 15000}])

```
> db.Teachers.insertMany([
...   {Tname: "Praveen", dno: 1, dname: "COMP", experience: 5, salary: 120
...   {Tname: "Amit", dno: 2, dname: "IT", experience: 3, salary: 9000},
...   {Tname: "Sneha", dno: 3, dname: "E&TC", experience: 7, salary: 15000
... ])
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("69cf963c6405d8c6816cb32c"),
                ObjectId("69cf963c6405d8c6816cb32d"),
                ObjectId("69cf963c6405d8c6816cb32e")
        ]
}
```

db.Students.insertMany([{Sname: "xyz", roll_no: 1, class: "SE"}, {Sname: "abc", roll_no: 2, class: "TE"}])

```
> db.Students.insertMany([
...   {Sname: "xyz", roll_no: 1, class: "SE"},
...   {Sname: "abc", roll_no: 2, class: "TE"},
...   {Sname: "pqr", roll_no: 3, class: "BE"}
... ])
{
        "acknowledged" : true,
        "insertedIds" : [
                ObjectId("69cf96a06405d8c6816cb32f"),
                ObjectId("69cf96a06405d8c6816cb330"),
                ObjectId("69cf96a06405d8c6816cb331")
        ]
}
```

3. Find the information about all teachers

db.Teachers.find()

```
> db.Teachers.find()
{ "_id" : ObjectId("69cf963c6405d8c6816cb32c"), "Tname" : "Praveen", "dno" : 1, "dname" : "COMP", "experience" : 5, "salary" : 12000 }
{ "_id" : ObjectId("69cf963c6405d8c6816cb32d"), "Tname" : "Amit", "dno" : 2, "dname" : "IT", "experience" : 3, "salary" : 9000 }
{ "_id" : ObjectId("69cf963c6405d8c6816cb32e"), "Tname" : "Sneha", "dno" : 3, "dname" : "E&TC", "experience" : 7, "salary" : 15000 }
```

4. Find the information about all teachers of computer department

db.Teachers.find({dname: "COMP"})

```
> db.Teachers.find({dname: "COMP"})
{ "_id" : ObjectId("69cf963c6405d8c6816cb32c"), "Tname" : "Praveen", "dno" : 1, "dname" : "COMP", "experience" : 5, "salary" : 12000
```

5. Find the information about all teachers of computer,IT,and e&TC department

db.Teachers.find({dname: {$in: ["COMP", "IT", "E&TC"]}})

```
> db.Teachers.find({dname: {$in: ["COMP", "IT", "E&TC"]}})
{ "_id" : ObjectId("69cf963c6405d8c6816cb32c"), "Tname" : "Praveen", "dno" : 1, "dname" : "COMP", "experience" : 5, db.Teachers.find({dname:
$in: ["COMP", "IT", "E&TC"]}, salary: {$gte: 10000}})
{ "_id" : ObjectId("69cf963c6405d8c6816cb32c"), "Tname" : "Praveen", "dno" : 1, "dname" : "COMP", "experience" : 5, "salary" : 12000 }
{ "_id" : ObjectId("69cf963c6405d8c6816cb32e"), "Tname" : "Sneha", "dno" : 3, "dname" : "E&TC", "experience" : 7, "salary" : 15000 }
```

6. Find the information about all teachers of computer,IT,and E&TC department having salary greater than or equl to 10000/-

db.Teachers.find({dname: {$in: ["COMP", "IT", "E&TC"]}, salary: {$gte: 10000}})

```
db.Teachers.find({dname: {
```

```
", "IT", "E&TC"]}, salary: {$gte: 10000}})
bjectId("69cf963c6405d8c6816cb32c"), "Tname" : "Praveen", "dno" : 1, "dname" : "COMP", "experience" : 5, "salary" : 12000 }
bjectId("69cf963c6405d8c6816cb32e"), "Tname" : "Sneha", "dno" : 3, "dname" : "E&TC", "experience" : 7, "salary" : 15000 }
```

7. Find the student information having roll_no = 2 or Sname=xyz

db.Students.find({$or: [{roll_no: 2}, {Sname: "xyz"}]})

```
> db.Students.find({$or: [{roll_no: 2}, {Sname: "xyz"}]})

{ "_id" : ObjectId("69cf96a06405d8c6816cb32f"), "Sname" : "xyz", "roll_no" : 1, "class" : "SE" }
{ "_id" : ObjectId("69cf96a06405d8c6816cb330"), "Sname" : "abc", "roll_no" : 2, "class" : "TE" }
```

8. Update the experience of teacher-praveen to 10years, if the entry is not available in database consider the entry as new entry.

db.Teachers.updateOne({Tname: "Praveen"}, {$set: {experience: 10}}, {upsert: true})

```
db.Teachers.updateOne(
..   {Tname: "Praveen"},
..   {$set: {experience: 10}},
..   {upsert: true}
.. )
"acknowledged" : true, "matchedCount" : 1, "modifiedCount" : 1 }
```

9. Update the deparment of all the teachers working in IT deprtment to COMP

db.Teachers.updateMany({dname: "IT"}, {$set: {dname: "COMP"}})

```
> db.Teachers.updateMany(
...   {dname: "IT"},
...   {$set: {dname: "COMP"}}
... )
{ "acknowledged" : true, "matchedCount" : 1, "modifiedCount" : 1 }
```

10. find the teachers name and their experience from teachers collection

db.Teachers.find({}, {Tname: 1, experience: 1, _id: 0})

```
> db.Teachers.find({}, {Tname: 1, experience: 1, _id: 0})

{ "Tname" : "Praveen", "experience" : 10 }
{ "Tname" : "Amit", "experience" : 3 }
{ "Tname" : "Sneha", "experience" : 7 }
```

11. Using Save() method insert one entry in department collection

db.Department.save({dno: 1, dname: "COMP"})

```
> db.Department.save({dno: 1, dname: "COMP"})
WriteResult({ "nInserted" : 1 })
```

12. Using Save() method change the dept of teacher praveen to IT

var doc = db.Teachers.findOne({Tname: "Praveen"}) doc.dname = "IT"
db.Teachers.save(doc)

```
> var doc = db.Teachers.findOne({Tname: "Praveen"})
> doc.dname = "IT"
IT
```

```
> db.Teachers.save(doc)
WriteResult({ "nMatched" : 1, "nUpserted" : 0, "nModified" : 1 })
```

13. Delete all the documents from teachers collection having IT dept.

db.Teachers.deleteMany({dname: "IT"})

```
> db.Teachers.deleteMany({dname: "IT"})
{ "acknowledged" : true, "deletedCount" : 1 }
```

14. display with pretty() method, the first 2 documents in teachers collection in ascending order

db.Teachers.find().sort({Tname: 1}).limit(2).pretty()

```
> db.Teachers.find().sort({Tname: 1}).limit(3).pretty()
{
        "_id" : ObjectId("69cf963c6405d8c6816cb32d"),
        "Tname" : "Amit",
        "dno" : 2,
        "dname" : "COMP",
        "experience" : 3,
        "salary" : 9000
}
{
        "_id" : ObjectId("69cf963c6405d8c6816cb32e"),
        "Tname" : "Sneha",
        "dno" : 3,
        "dname" : "E&TC",
        "experience" : 7,
        "salary" : 15000
}
```